

Performance Evaluation of 802.11 Adaptive Modulation and Error Rates using Log-Distance Path Loss Modeling

1. Abstract

In this project, the performance of 802.11 wireless local area networks (WLANs) was evaluated by simulating physical layer (PHY) signal degradation over distance. Using MATLAB, a mathematical model was developed to compare 2.4 GHz and 5 GHz frequency bands in an indoor environment (Path Loss Exponent $n=3$), with the effect of a physical concrete wall obstacle. The simulation calculates the Signal-to-Noise Ratio (SNR), maps it to dynamic throughput using an Adaptive Modulation and Coding Scheme (MCS), and computes the theoretical Symbol Error Rate (SER). The results visually demonstrate distance, frequency, and throughput, proving that adaptive modulation is critical for maintaining connection integrity at the network edge.

2. Literature Review

The performance evaluation of wireless networks fundamentally relies on modeling the physical environment and the system's response to signal degradation. This simulation was built upon three core telecommunications concepts established in wireless literature:

2.1 Signal Propagation and Log-Distance Path Loss

In indoor environments, free-space models are insufficient due to multipath fading, reflection, and physical obstacles. The Log-Distance Path Loss model is widely utilized to predict signal attenuation inside buildings. As established in wireless communications theory, adjusting the path loss exponent (n) allows for accurate modeling of dense environments. For this simulation, an exponent of $n=3$ was selected to represent a standard academic or office building, coupled with a static attenuation penalty to simulate physical walls.

2.2 Symbol Error Rate (SER) in AWGN Channels

Evaluating connection reliability requires understanding how noise affects symbol decoding. Literature defines the Additive White Gaussian Noise (AWGN) channel model as the standard baseline for calculating theoretical error limits. As the Signal-to-Noise Ratio (SNR) decreases, the distance between symbols in the I/Q constellation diagram shrinks relative to the noise floor, causing an exponential increase in the Symbol Error Rate using standard Q-function and complementary error function (erfc) mathematics.

2.3 IEEE 802.11 Adaptive Modulation and Coding (AMC)

The IEEE 802.11 standards dictate that devices must dynamically adjust their Modulation and Coding Scheme (MCS) based on current channel conditions. Academic literature highlights AMC as the primary mechanism for balancing throughput with reliability. High-density modulations like 64-QAM offer high data rates but require high SNR. When the SER exceeds acceptable thresholds, the 802.11 MAC layer mandates a downshift to more robust schemes like 16-QAM or BPSK to maintain link stability, which forms the core "staircase" behavior modeled in this report.

3. System Model and Methodology

3.1 Log-Distance Path Loss Model

To simulate a realistic indoor environment (such as a university campus building), the Log-Distance Path Loss (PL) model was employed. The total path loss is calculated as:

$$PL(d) = L_0 + 10n\log_{10}(d) + X_w$$

Where:

- L_0 is the Free Space Path Loss at a 1-meter reference distance: $L_0 = 20\log_{10}(1) + 20\log_{10}(f) - 147.55$
- n is the path loss exponent.
- X_w is the cost or degradation of a signal applied when the signal passes through a physical obstacle.

3.2 Signal-to-Noise Ratio (SNR)

The SNR (in dB) is determined by subtracting the total path loss from the transmission power (P_t) and comparing it against the system noise floor (N_f):

$$SNR_{db} = (P_t - PL(d)) - N_f$$

3.3 Adaptive Modulation and Coding Scheme (MCS)

The system continuously monitors the SNR. As it drops below critical thresholds, the modulation order (M) is reduced to preserve the connection. The simulated thresholds are:

- **SNR > 25 dB:** 64-QAM (54 Mbps)
- **SNR > 20 dB:** 16-QAM (36 Mbps)
- **SNR > 15 dB:** QPSK / 4-QAM (24 Mbps)
- **SNR > 5 dB:** BPSK (6 - 12 Mbps)

- **SNR < 5 dB:** Disconnected (0 Mbps)

3.4 Symbol Error Rate (SER) Calculation

To prove the necessity of the MCS downshifting, the theoretical SER was calculated over an Additive White Gaussian Noise (AWGN) channel. The SNR is first converted to a linear scale (γ_8).

For BPSK ($M=2$):

$$P_8 = \frac{1}{2} \text{erfc}(\sqrt{\gamma_8})$$

For M-QAM ($M > 2$):

$$P_8 \approx 4(1 - \frac{1}{\sqrt{M}})Q(\sqrt{\frac{3\gamma_8}{M-1}})$$

Where:

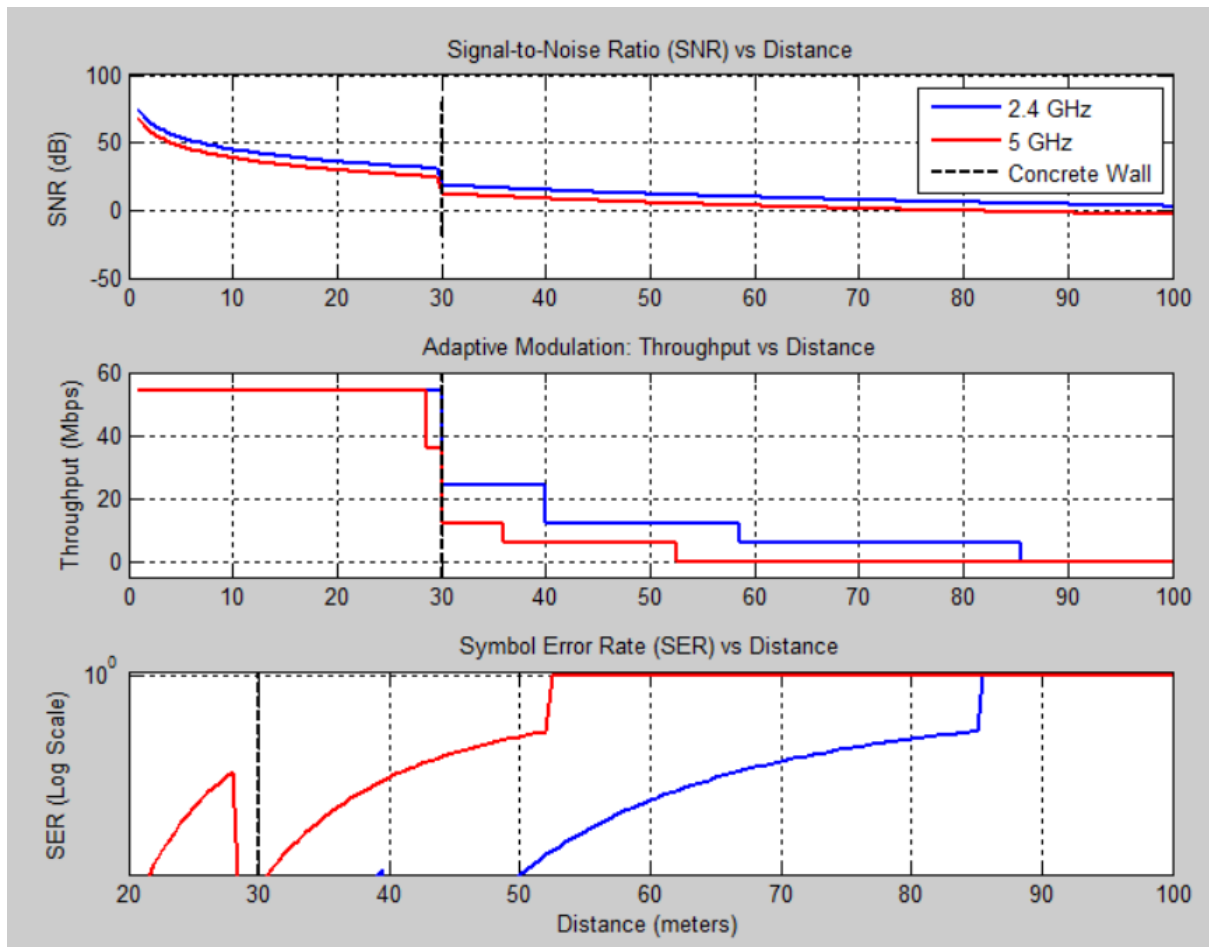
- **M** is the modulation order (e.g., 16 for 16-QAM)

4. Simulation Parameters

Using MATLAB and the following constants, chosen to represent standard commercial Wi-Fi hardware operating in a dense indoor environment:

Parameter	Value
Frequencies (f)	2.4 GHz and 5.0 GHz
Transmit Power (P_t)	20 dBm
Noise Floor (N_f)	-95 dBm
Path Loss Exponent (n)	3.0 (Indoor environment)
Obstacle Location (Concrete Wall)	30 meters
Obstacle Attenuation (X_w)	12 dB (Concrete)
Simulation Distance	1 to 100 meters

5. Results



5.1 Signal-to-Noise Ratio Analysis

The top plot of the simulation illustrates the SNR decay for both 2.4 GHz and 5 GHz bands. Due to its higher frequency, the 5 GHz signal experiences a higher initial Free Space Path Loss (L_0), causing its curve to remain permanently lower than the 2.4 GHz curve. At $d = 30$ meters, the introduction of a concrete wall induces an immediate, sharp ~ 12 dB drop in both signals, demonstrating the vulnerability of wireless signals to physical obstructions.

5.2 Throughput and the "Staircase" Pattern

In the middle plot we can see the discrete nature of digital modulation. Rather than a smooth curve, the throughput follows a "staircase" pattern. As the station moves away and SNR degrades, the hardware determines that the current modulation is no longer viable. For instance, the 5 GHz band is forced to drop from 54 Mbps to 36 Mbps much earlier than the 2.4 GHz band. Furthermore, the 12 dB penalty at the concrete wall causes a throughput drop for the 5 GHz signal, crashing it to the minimum 6 Mbps rate, whereas the 2.4 GHz signal maintains a 24 Mbps connection through the barrier.

5.3 Symbol Error Rate (SER) and the "Sawtooth" Pattern

The bottom plot reveals logic of the Media Access Control (MAC) layer. Plotted on a logarithmic scale, the SER climbs exponentially as distance increases within a specific modulation tier. Left unchecked, this error rate would sever the connection. However, exactly as the SER approaches critical failure levels, the system downshifts the modulation (e.g., from 64-QAM to 16-QAM). This action instantly resets the error rate to a highly reliable state (e.g., dropping from 10^{-2} to 10^{-6}), creating a distinct "sawtooth" visual pattern. This proves that throughput reductions are actively engineered to preserve data integrity.

6. Conclusion

This simulation successfully modeled the physical layer behaviors of 802.11 WLANs. The mathematical evaluation confirms two primary networking principles. First, while 5 GHz offers a cleaner channel with potentially less interference, it sacrifices physical penetration and range compared to 2.4 GHz. Second, adaptive modulation is a mandatory feature for mobile wireless networks, by intelligently sacrificing data rate (throughput), the system successfully mitigates exponential error growth, maintaining a stable connection at the extreme edges of the network coverage area.